



THE OHIO STATE UNIVERSITY

FISHER COLLEGE OF BUSINESS

The Stochastic Discount Factor

(BUSFIN 8200 - Asset Pricing Foundations)

Andrei S. Gonçalves

This Module (The Stochastic Discount Factor)

(1) The SDF

[1.1] The SDF Characterization $1 = \mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}]$

[1.2] The SDF & Risk Adjustments $\mathbb{E}_t[R_j] = \mathbb{E}_t[R_0] + \text{Cov}_t\left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1}\right]$

(2) General SDF Estimation

[2.1] The Generalized Method of Moments (GMM)

$$\mathbb{E}[\epsilon(\theta)] = \mathbf{0} \Rightarrow \hat{A} \hat{\mathbb{E}}[\epsilon(\hat{\theta})] = \mathbf{0}$$

[2.2] Estimating the SDF by GMM

$$\mathbb{E}_t[M(F_{t+1}, \theta) \cdot R_{t+1}] = 1 \Rightarrow \hat{A} \mathbb{E}[M(F_t, \hat{\theta}) \cdot R_t - 1] = \mathbf{0}$$

(3) Linear & Log-Linear SDF Models

[3.1] Linear SDF & Covariance Pricing Models

$$M_{t+1} = \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1}) \Rightarrow \mathbb{E}_t[xR_j] = b_t' \text{Cov}_t[F, xR_j]$$

[3.2] Linear SDF & Beta Pricing Models

$$M_{t+1} = \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1}) \Rightarrow \mathbb{E}_t[xR_j] = \lambda_t' \beta_{j,t}$$

[3.2] Log-Linear SDF Models

$$m_{t+1} = a_t - b_t' f_{t+1} \Rightarrow \mathbb{E}[x r_j] + 0.5 \cdot (\text{Var}[r_j] - \text{Var}[\bar{r}_j]) = \lambda_t' \beta_{j,t}$$

Outline

(1) The SDF

[1.1] The SDF Characterization

[1.2] The SDF & Risk Adjustments

(2) General SDF Estimation

[2.1] The Generalized Method of Moments (GMM)

[2.2] Estimating the SDF by GMM

(3) Linear & Log-Linear SDF Models

[3.1] Linear SDF & Covariance Pricing Models

[3.2] Linear SDF & Beta Pricing Models

[3.3] Log-Linear SDF Models

(1) The SDF

- We saw that much of the variation in asset prices is due to variation in $\mathbb{E}[R]$
- As such, this section builds a general theory for why $\mathbb{E}[R]$ varies
- The theory is based on the concept of a Stochastic Discount Factor (SDF)

[1.1] The SDF Characterization

- An SDF is a random variable, M , that satisfies $P_{j,t} = \mathbb{E}_t[M_{t+1} \cdot X_{j,t+1}]$
- If $X = R$, then $1 = \mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}]$
- Economically, $\uparrow M \Leftrightarrow \uparrow$ Marginal Utility
- So a payoff (or return) is more valuable if it is received at a high M state
- We will see that (under No-Arbitrage) a $M > 0$ always exists
- Moreover, there is an M that is linear in the maximum Sharpe ratio portfolio, R^*

[1.2] The SDF & Risk Adjustments

- $1 = \mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}]$ implies

$$\mathbb{E}_t[R_j] = \underbrace{\mathbb{E}_t[R_0]}_{\mathbb{E}[R] \text{ from Time Discount}} + \underbrace{\text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1} \right]}_{\mathbb{E}[R] \text{ Risk Adjustment}}$$

- We can write this equation as $\mathbb{E}_t[xR_j] = \beta_{j,t} \cdot \lambda_t$

* $\beta_{j,t} = \text{Cov}_t[-M, xR_j] / \text{Var}_t[M]$ is the asset j “risk exposure”

* $\lambda_t = \text{Var}_t[M] / \mathbb{E}_t[M]$ is the “risk premium” per unit of risk exposure

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The SDF in a Simple Two-Period Economy

- Consider an investor with wealth W_t in a two period economy (t and $t + 1$)
- At time t , the investor:
 - buys N_t units of a financial asset with price P_t
 - consumes the rest of his/her endowed wealth, $C_t = W_t - N_t \cdot P_t$
- At time $t + 1$, the investor
 - receives N_t units of the financial asset cash flow X_{t+1}
 - consumes all of his/her proceeds, $C_{t+1} = N_t \cdot X_{t+1}$
- The investor's problem is

$$\begin{aligned} \underset{\{C_t, C_{t+1}, N_t\}}{\text{Max}} \quad & U(C_t) + \delta \cdot \mathbb{E}_t[U(C_{t+1})] \quad \text{s.t.} \quad C_t = W_t - N_t \cdot P_t \\ & C_{t+1} = N_t \cdot X_{t+1} \end{aligned}$$

- Substituting the budget constraints and taking derivative w.r.t N_t yields

$$P_t = \mathbb{E}_t \left[\delta \cdot \frac{\partial U(C_{t+1})}{\partial U(C_t)} \cdot X_{t+1} \right]$$

- Defining the **Stochastic Discount Factor (SDF)** $M_{t+1} = \delta \cdot \frac{\partial U(C_{t+1})}{\partial U(C_t)}$, we have

$$P_t = \mathbb{E}_t [M_{t+1} \cdot X_{t+1}]$$

- Dividing both sides of the equation by P_t , we have (w/ $R_{t+1} = X_{t+1}/P_t$)

$$1 = \mathbb{E}_t [M_{t+1} \cdot R_{t+1}]$$

The SDF in Complete Markets

- There are $n = 1, 2, \dots, N$ states of nature with probability $\pi(n) > 0$
- **Law of One Price (LOP):**
 - Two assets with identical payoffs in every state of nature must have the same price
- **No-Arbitrage:**
 - An asset with **Payoff** ≥ 0 must have **Price** ≥ 0
 - An asset with **Payoff** ≥ 0 and a chance for **Payoff** > 0 must have **Price** > 0
- A “contingent claim” paying \$1 if state of nature n materializes has price $P_c(n)$
(also called Arrow-Debreu securities due to Debreu (1959) and Arrow (1971))
- **Complete Markets:** there is one contingent claim for each state of nature n
- Complete Markets + LOP $\Leftrightarrow$ There is a unique SDF

- Asset with payoff $X(n)$ is worth

$$P = \sum_{n=1}^N P_c(n) \cdot X(n) = \sum_{n=1}^N \pi(n) \cdot \underbrace{\frac{P_c(n)}{\pi(n)}}_{M(n)} \cdot X(n)$$

- This leads to the “fundamental equation of asset pricing” (or the SDF equation)

$$P_t = \mathbb{E}_t[M_{t+1} \cdot X_{t+1}]$$

- With $P_t = 1$ and $X_{t+1} = R_{t+1}$:

$$1 = \mathbb{E}_t[M_{t+1} \cdot R_{t+1}]$$

and

$$0 = \mathbb{E}_t[M_{t+1} \cdot xR_{t+1}]$$

- Complete Markets + No-Arbitrage $\Leftrightarrow$ There is a unique positive SDF

- $M(n) > 0$ because $P_c(n) > 0$ (since **Payoff** = \$1 and $\pi(n) > 0$)

The SDF in Incomplete Markets

- There are $n = 1, 2, \dots, N$ states of nature with probability $\pi(n) > 0$
- **Law of One Price (LOP):**
 - Two assets with identical payoffs in every state of nature must have the same price
- **No-Arbitrage:**
 - An asset with **Payoff** ≥ 0 must have **Price** ≥ 0
 - An asset with **Payoff** ≥ 0 and a chance for **Payoff** > 0 must have **Price** > 0
- **Incomplete Markets:** we have $j = 1, 2, \dots, J < N$ non-redundant assets
- **Incomplete Markets + LOP** $\Leftrightarrow$ There is a unique tradable SDF
 - $X = [X_1 \ X_2 \ \dots \ X_J]'$ and $P = [P_1 \ P_2 \ \dots \ P_J]'$
 - A tradable SDF must look like $M_{t+1}^o = b_t' X_{t+1}$ (it costs $b_t' P_t$)
 - M^o must price all assets: $P_t = \mathbb{E}_t[M^o \cdot X] = \mathbb{E}_t[X \cdot (M^o)'] = \mathbb{E}_t[XX'] b_t$
 - Solving for b_t : $b_t = \mathbb{E}_t[XX']^{-1} P_t \Rightarrow M_{t+1}^o = P_t' \mathbb{E}_t[XX']^{-1} X_{t+1}$
 - LOP ensures that $\mathbb{E}_t[XX']^{-1}$ exists and is unique
 - With $P = \mathbf{1}$ & $X = R$: $b_t = \mathbb{E}_t[RR']^{-1} \mathbf{1}$ & $M_{t+1}^o = \mathbf{1}' \mathbb{E}_t[RR']^{-1} R_{t+1}$
 - We have ∞ non-tradable SDFs: $M = M^o + \epsilon$ s.t. $\mathbb{E}_t[\epsilon \cdot X] = 0$
- **Incomplete Markets + No-Arbitrage** $\Leftrightarrow$ There is a positive SDF
 - The positive SDF does not have to be our tradable M^o
 - In complete markets, the SDF is unique, positive, and tradable

The SDF & The Max Sharpe Ratio Portfolio

- Same setup as in the prior slide
 - $n = 1, 2, \dots, N$ states of nature
 - $j = 1, 2, \dots, J < N$ non-redundant assets
- Let's construct an SDF that is linear in **Payoffs** ► (Other Useful SDF)

◦ We want $M_{t+1}^* = \mathbb{E}_t[M^*] \cdot (1 - b_t'(X_{t+1} - \mathbb{E}_t[X]))$

◦ M^* must price all assets:

$$P_t = \mathbb{E}_t[M^* \cdot X] = \mathbb{E}_t[X \cdot (M^*)'] = \mathbb{E}_t[M^*] \cdot (\mathbb{E}_t[X] - \underbrace{\mathbb{E}_t[X(X - \mathbb{E}_t[X])']}_{\Sigma_t} b_t)$$

◦ Solving for b_t :

$$b_t = \Sigma_t^{-1} (\mathbb{E}_t[X] - P_t / \mathbb{E}_t[M^*])$$

◦ If risk-free payoff ($X_{f,t+1} = 1$) exists, then M^* is our tradable M^0

◦ If risk-free payoff ($X_{f,t+1} = 1$) does not exist, then M^* is a non-tradable **SDF**

- Consider the special case with $P = 0$ & $X = xR$:

$$M_{t+1}^* = \mathbb{E}_t[M^*] \cdot (a_t - b_t' xR_{t+1})$$

where

$$a_t = 1 + b_t' \mathbb{E}_t[xR] \quad \& \quad b_t = \Sigma_t^{-1} \mathbb{E}_t[xR]$$

◦ Note that $b_t = \operatorname{argmax} \left(\operatorname{SR}_t[xR^*] = \mathbb{E}_t[b_t' xR] / \operatorname{Vol}_t[b_t' xR] \right) = \Sigma_t^{-1} \mathbb{E}_t[xR]$

◦ So, $xR_{t+1}^* = b_t' xR_{t+1}$ is the max Sharpe Ratio portfolio

◦ And $M_{t+1}^* = \mathbb{E}_t[M^*] \cdot (a_t - xR_{t+1}^*)$ is linear in the max Sharpe Ratio portfolio **1.5**

$\mathbb{E}[\text{SDF}]$ Relation to the Zero-Beta and Risk-Free Returns

- The zero-beta return (R_0) for SDF M is the portfolio that solves the problem

$$\begin{aligned} \underset{\omega_{0,t}}{\text{Min}} \quad & \left(\mathbb{V}ar_t[R_{0,t+1}] = \mathbb{V}ar_t[\omega_{0,t}' R_{t+1}] \right) \quad \text{s.t.} \quad \omega_{0,t}' \mathbf{1} = 1 \\ & \text{Cov}_t[R_0, M] = 0 \end{aligned}$$

- This problem has a unique solution
- We call R_0 “the zero-beta return” for simplicity (we do not index to M)
- But each SDF has its own R_0 (the relevant SDF should be clear from the context)
- R_0 has an important property:

$$0 = \text{Cov}_t[R_0, M] = \underbrace{\mathbb{E}_t[R_0 \cdot M]}_1 - \mathbb{E}_t[R_0] \cdot \mathbb{E}_t[M] \Rightarrow \boxed{\mathbb{E}_t[M] = 1/\mathbb{E}_t[R_0]}$$

- If a risk-free return (R_f) exists, then $R_f = R_0$ (for any M)
 - The risk-free payoff is $X_{f,t+1} = 1$ and it has price $P_{f,t} = \mathbb{E}_t[M \cdot X_f] = \mathbb{E}_t[M]$
 - So, the risk-free return at $t+1$ is $R_{f,t} = X_{f,t+1}/P_{f,t} = 1/\mathbb{E}_t[M]$
 - Note that the risk-free return at $t+1$ is known at t , and thus we write it as $R_{f,t}$
 - So, if a risk-free return exists, we have the property

$$\boxed{\mathbb{E}_t[M] = 1/R_{f,t}}$$

- In fact, since $\mathbb{V}ar_t[R_f] = \text{Cov}_t[R_f, M] = 0$, we have $R_{f,t} = R_{0,t+1} = \mathbb{E}_t[R_0]$
- Note: in complete markets, R_f always exists ($P_f = \sum_{n=1}^N P_c(n)$)

SDF Relation to Risk-Neutral Pricing

- Risk-Neutral pricing logic: $P_t = \mathbb{E}_t[M_{t+1} \cdot X_{t+1}]$

$$\begin{aligned} &= \sum_{n=1}^N \pi(n) \cdot M(n) \cdot X(n) \\ &= \frac{1}{\mathbb{E}_t[R_0]} \cdot \sum_{n=1}^N \underbrace{\pi(n) \cdot \frac{M(n)}{\mathbb{E}_t[M]}}_{\pi^{RN}(n)} \cdot X(n) \end{aligned}$$

- If we use a $M > 0$ (i.e., assume No-Arbitrage), then we have

$$* \pi^{RN}(n) > 0 \quad (M(n) > 0 \text{ and } \pi(n) > 0)$$

$$* \sum_{n=1}^N \pi^{RN}(n) = 1 \quad (\text{since } \mathbb{E}_t[M] = \sum_{n=1}^N \pi(n) \cdot M(n))$$

- * So, $\pi^{RN}(n)$ is a probability measure (the risk-neutral probability measure)

$$P_t = \mathbb{E}_t[M \cdot X] = \frac{1}{\mathbb{E}_t[R_0]} \cdot \mathbb{E}_t^{RN}[X]$$

- Risk-Neutral pricing in complete markets:

- In this case, a risk-free return exists, so

$$P_t = \mathbb{E}_t[M \cdot X] = \frac{1}{R_{f,t}} \cdot \mathbb{E}_t^{RN}[X]$$

- Moreover, the $\pi^{RN}(n)$ probabilities only depend on the contingent claims

$$\pi^{RN}(n) = \pi(n) \cdot \frac{M(n)}{\mathbb{E}_t[M]} = \pi(n) \cdot \frac{P_c(n)/\pi(n)}{1/R_{f,t}} = R_{f,t} \cdot P_c(n) = \frac{P_c(n)}{\sum_{n=1}^N P_c(n)}$$

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Risk Adjustment in Asset Prices

- Asset prices embed a risk adjustment:

$$\begin{aligned}
 P_{j,t} &= \mathbb{E}_t[M_{t+1} \cdot X_{j,t+1}] \\
 &= \mathbb{E}_t[M_{t+1}] \cdot \mathbb{E}_t[X_{j,t+1}] + \text{Cov}_t[M_{t+1}, X_{j,t+1}] \\
 &= \underbrace{\frac{1}{\mathbb{E}_t[R_0]} \cdot \mathbb{E}_t[X_{j,t+1}]}_{\text{Time Discounted Payoff}} + \underbrace{\text{Cov}_t[M_{t+1}, X_{j,t+1}]}_{\text{Payoff Risk Adjustment}}
 \end{aligned}$$

- Assets with **payoffs** that comove positively with the **SDF** have higher value

- Intuition:

* Consider $M_{t+1} = \delta \cdot \partial U(C_{t+1}) / \partial U(C_t)$

* If $X_{j,t+1}$ tends to be high when $\partial U(C_{t+1})$ is high, then the asset is valuable

* It gives the investor more money when money is most valuable ("bad times")

- The risk adjustment on prices is really about all future cash flows:

- If $X_{j,t+1} = CF_{j,t+1} + P_{j,t+1}$, then $P_{j,t} = \mathbb{E}_t[M_{t+1} \cdot (CF_{j,t+1} + P_{j,t+1})]$

- Assuming $\mathbb{E}_t[M_{t \rightarrow t+h} \cdot P_{j,t+h}] \xrightarrow{h \rightarrow \infty} 0$, recursively substituting $P_{j,t+h}$ yields

$$\begin{aligned}
 P_{j,t} &= \sum_{h=1}^{\infty} \mathbb{E}_t[M_{t \rightarrow t+h} \cdot CF_{j,t+h}] \\
 &= \underbrace{\sum_{h=1}^{\infty} \frac{1}{\mathbb{E}_t[R_{0,t \rightarrow t+h}]} \cdot \mathbb{E}_t[CF_{j,t+h}]}_{\text{Time Discounted Cash Flows}} + \underbrace{\sum_{h=1}^{\infty} \text{Cov}_t[M_{t \rightarrow t+h}, CF_{j,t+h}]}_{\text{Cash Flow Risk Adjustment}}
 \end{aligned}$$

w/ $M_{t \rightarrow t+h} = \prod_{\tau=1}^h M_{t+\tau}$ and $\mathbb{E}_t[M_{t \rightarrow t+h}] = 1 / \mathbb{E}_t[R_{0,t \rightarrow t+h}]$

Risk Adjustment in Expected Returns

- Expected Returns embed a risk adjustment:

$$\begin{aligned}
 1 &= \mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}] \\
 &= \mathbb{E}_t[M_{t+1}] \cdot \mathbb{E}_t[R_{j,t+1}] + \text{Cov}_t[M_{t+1}, R_{j,t+1}] \\
 &\Downarrow \\
 \mathbb{E}_t[R_j] &= \underbrace{\mathbb{E}_t[R_0]}_{\mathbb{E}[R] \text{ from Time Discount}} + \underbrace{\text{Cov}_t\left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1}\right]}_{\mathbb{E}[R] \text{ Risk Adjustment}}
 \end{aligned}$$

- Assets with returns that comove negatively with the SDF have higher $\mathbb{E}[R]$
 - * Consider $M_{t+1} = \delta \cdot \partial U(C_{t+1}) / \partial U(C_t)$
 - * If R_{t+1} tends to be low when $\partial U(C_{t+1})$ is high, then the asset is “risky”
 - * It provides low returns when money is most valuable (“bad times”)
 - * Investors require a compensation (a “risk premium”) to hold these assets
- When testing models, it is often useful to consider expected log returns
 - Assume (M, R) are conditionally log-Normal
 - This implies (m, r) are conditionally Normal
 - In this case, the expected return equation can be written as ► (Proof)

$$\underbrace{\mathbb{E}_t[r_j] + 0.5 \cdot \text{Var}_t[r_j]}_{\log(\mathbb{E}_t[R_j])} = \underbrace{\mathbb{E}_t[r_0] + 0.5 \cdot \text{Var}_t[r_0]}_{\log(\mathbb{E}_t[R_0])} + \underbrace{\text{Cov}_t[-m_{t+1}, r_{j,t+1}]}_{\mathbb{E}[r] \text{ Risk Adjustment}}$$

- * If a risk-free return exists, then $\mathbb{E}_t[r_0] + 0.5 \cdot \text{Var}_t[r_0] = r_{f,t} = \log(R_{f,t})$

The Risk Premium Equation

$$\mathbb{E}_t[R_j] = \mathbb{E}_t[R_0] + \text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1} \right]$$

- To avoid measuring R_0 , much empirical work is done with excess returns
- If xR_j represents the excess return relative to some arbitrary asset, then

$$\mathbb{E}_t[xR_j] = \text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, xR_{j,t+1} \right] = \text{Cov}_t [xR_{t+1}^*, xR_{j,t+1}]$$

- We can then write this risk premium equation as

$$\mathbb{E}_t[xR_j] = \beta_{j,t} \cdot \lambda_t$$

where

- * $\beta_{j,t} = \text{Cov}_t[-M, xR_j] / \text{Var}_t[M]$ is the asset j “risk exposure”
- * $\lambda_t = \text{Var}_t[M] / \mathbb{E}_t[M]$ is the “risk premium” per unit of risk exposure
- Or we can work with a risk premium equation with a tradable “risk factor” (xR^*)

$$\mathbb{E}_t[xR_j] = \beta_{j,t} \cdot \lambda_t$$

where

- * $\beta_{j,t} = \text{Cov}_t[xR^*, xR_j] / \text{Var}_t[xR^*]$ is the asset j “risk exposure”
- * $\lambda_t = \text{Var}_t[xR^*] = \mathbb{E}_t[xR^*]$ is the “risk premium” per unit of risk exposure
- Time variation in aggregate $\mathbb{E}_t[xR]$ is linked to time variation in β_t or λ_t
- Cross-asset variation in $\mathbb{E}_t[xR_j]$ is linked to cross-asset variation in $\beta_{j,t}$

Lower Bounds for the SDF Volatility

- Our risk premium equation is:

$$\begin{aligned}\mathbb{E}_t[xR_j] &= \text{Cov}_t \left[-\frac{M}{\mathbb{E}_t[M]}, xR_j \right] \\ &= \frac{\text{Cor}_t[-M, xR_j] \cdot \text{Vol}_t[M] \cdot \text{Vol}_t[xR_j]}{\mathbb{E}_t[M]}\end{aligned}$$

⇓

$$\text{SR}_t[xR_j] = \frac{\mathbb{E}_t[xR_j]}{\text{Vol}_t[xR_j]} = \text{Cor}_t[-M, xR_j] \cdot \frac{\text{Vol}_t[M]}{\mathbb{E}_t[M]}$$

- Since $|\text{Cor}_t[-M, xR_j]| \leq 1$, we have a lower bound for the SDF volatility:

$$|\text{SR}_t[xR_j]| = \frac{|\mathbb{E}_t[xR_j]|}{\text{Vol}_t[xR_j]} \leq \frac{\text{Vol}_t[M]}{\mathbb{E}_t[M]}$$

- The US equity Sharpe ratio was in the range of 0.33 to 0.50 in the 20th century
- Since $\mathbb{E}_t[M] = 1/\mathbb{E}_t[R_0] \approx 1$, this means the SDF must be very volatile
- We have a much sharper bound when using $xR^* = b_t' xR = (\Sigma_t^{-1} \mathbb{E}_t[xR])' xR$
 - To start, note that $\mathbb{E}_t[xR^*] = \text{Var}_t[xR^*] = \mathbb{E}_t[xR'] \Sigma_t^{-1} \mathbb{E}_t[xR] > 0$
 - As such, the prior bound with $xR_j = xR^*$ yields (Hansen, Jagannathan (1991))

$$\text{SR}_t[xR^*] = \frac{\sqrt{\mathbb{E}_t[xR'] \Sigma_t^{-1} \mathbb{E}_t[xR]}}{\text{Vol}_t[M]} \leq \frac{\text{Vol}_t[M]}{\mathbb{E}_t[M]}$$

- This is known as the Hansen-Jagannathan bound (w/ $X = xR$ and $P = 0$)

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(2) General SDF Estimation

- We now have a general theory to explain variation in expected returns:

$$\mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}] = 1 \Rightarrow \mathbb{E}_t[R_j] = \mathbb{E}_t[R_0] + \text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1} \right]$$

- Each economic model we write leads to a different SDF, $M_t = M(F_t, \theta)$
- The SDF is a function of data (F_t) and parameters (θ)
- We need a method to estimate θ

[2.1] The Generalized Method of Moments (GMM)

- Many statistical models can be written as “moment conditions” $\mathbb{E}[\epsilon(\theta)] = 0$
- The GMM uses sample averages to replace population moments
- We can have more moment conditions than parameters, so we use $\hat{A} \hat{\mathbb{E}}[\epsilon(\hat{\theta})] = 0$
- If we minimize $\hat{\mathbb{E}}[\epsilon(\hat{\theta})]' \hat{W} \hat{\mathbb{E}}[\epsilon(\hat{\theta})]$, then we have $\hat{A} = \hat{G}' \hat{W}$ (w/ $\hat{G} = \partial_{\theta} \hat{\mathbb{E}}[\epsilon(\hat{\theta})]$)
- Using $\hat{W} = \hat{S}^{-1}$ results in the efficient GMM (w/ $\hat{S} = \sum_{\ell=-\infty}^{\infty} \mathbb{E}[\epsilon_t(\theta) \epsilon_{t-\ell}(\theta)']$)

[2.2] Estimating the SDF by GMM

- The SDF equation is $\mathbb{E}_t[M(F_{t+1}, \theta) \cdot R_{t+1}] = 1$
- The law of iterated expectations then implies $\mathbb{E}[M(F_t, \theta) \cdot R_t] = 1$
- So, we can define $\epsilon(\theta) = M(F_t, \theta) \cdot R_t - 1$ and use GMM to estimate θ
- We have one moment condition for each asset included in R_t

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(2) General SDF Estimation

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Introducing the Method of Moments (MM)

- Consider the estimation of the mean of a random variable
 - We observe a sample of x_t ($t = 1, 2, \dots, T$)
 - We have $\mu = \mathbb{E}[x_t]$ and want to estimate the parameter μ
 - $\mu = \mathbb{E}[x_t] \Rightarrow \mathbb{E}[\epsilon(\mu)] = 0$, where $\epsilon(\mu) = f(x_t, \mu) = \mu - x_t$
 - The method of moments replaces $\mathbb{E}[\epsilon(\mu)]$ with $\widehat{\mathbb{E}}[\epsilon(\widehat{\mu})]$

$$\begin{aligned}
 0 &= \widehat{\mathbb{E}}[\epsilon(\widehat{\mu})] = \widehat{\mathbb{E}}[\widehat{\mu} - x_t] = \frac{1}{T} \cdot \sum_{t=1}^T (\widehat{\mu} - x_t) = \widehat{\mu} - \frac{1}{T} \cdot \sum_{t=1}^T x_t \\
 &\Downarrow \\
 \widehat{\mu} &= \frac{1}{T} \cdot \sum_{t=1}^T x_t
 \end{aligned}$$

- The method of moments can also be used to estimate a vector of parameters
 - We observe a sample of x_t ($t = 1, 2, \dots, T$)
 - We have $\mu = \mathbb{E}[x_t]$ & $\sigma^2 = \mathbb{E}[(x_t - \mu)^2]$ and want to estimate $\theta = (\mu, \sigma^2)$
 - This structure implies $\mathbb{E}[\epsilon(\theta)] = \mathbf{0}$, where $\epsilon(\theta) = \begin{pmatrix} \mu - x_t \\ \sigma^2 - (x_t - \mu)^2 \end{pmatrix}$
 - Setting $\boxed{\widehat{\mathbb{E}}[\epsilon(\widehat{\theta})] = \mathbf{0}}$:

$$\begin{aligned}
 \mathbf{0} &= \widehat{\mathbb{E}}[\epsilon(\widehat{\theta})] = \begin{pmatrix} \widehat{\mathbb{E}}[\widehat{\mu} - x_t] \\ \widehat{\mathbb{E}}[\widehat{\sigma}^2 - (x_t - \widehat{\mu})^2] \end{pmatrix} = \begin{pmatrix} \frac{1}{T} \cdot \sum_{t=1}^T (\widehat{\mu} - x_t) \\ \frac{1}{T} \cdot \sum_{t=1}^T (\widehat{\sigma}^2 - (x_t - \widehat{\mu})^2) \end{pmatrix} \\
 \begin{pmatrix} \widehat{\mu} \\ \widehat{\sigma}^2 \end{pmatrix} &\Downarrow \\
 &= \begin{pmatrix} \frac{1}{T} \cdot \sum_{t=1}^T x_t \\ \frac{1}{T} \cdot \sum_{t=1}^T (x_t - \widehat{\mu})^2 \end{pmatrix}
 \end{aligned}$$

The Generalized Method of Moments (GMM)

- The GMM generalizes the logic of the MM (Hansen (1982))
 - We observe a sample of $\mathbf{x}_t$ ($t = 1, 2, \dots, T$) with $\mathbf{x}_t$ being a vector
 - We have K parameters in θ
 - We have $Q \geq K$ moment conditions in $\boxed{\mathbb{E}[\epsilon(\theta)] = \mathbf{0}}$, where $\epsilon_t(\theta) = f(\mathbf{x}_t, \theta)$
 - The GMM estimate for θ solves the system of equations

$$\hat{\mathbf{A}} \hat{\mathbb{E}}[\epsilon(\hat{\theta})] = \mathbf{0}$$

- $\hat{\mathbf{A}} \xrightarrow{P} \mathbf{A}$ is a $K \times Q$ matrix defining K linear combinations of $\hat{\mathbb{E}}[\epsilon(\hat{\theta})]$ to set to zero
 - $\hat{\mathbf{A}}$ can be chosen to ensure that a $\hat{\theta}$ exists (under some regularity conditions)
 - If $Q = K$ then the system is “just identified” (and GMM = MM if $\hat{\mathbf{A}} = \mathbf{A} = \mathbf{I}$)
 - If $Q > K$ then the system is “overidentified” (and GMM $\neq$ MM for any $\hat{\mathbf{A}}$)
- The GMM estimate and errors have asymptotic normal distributions

$$\hat{\theta} \stackrel{T \rightarrow \infty}{\sim} \mathcal{N}(\theta, a\mathbb{V}ar[\hat{\theta}])$$

$$\hat{\mathbb{E}}[\epsilon(\hat{\theta})] \stackrel{T \rightarrow \infty}{\sim} \mathcal{N}(\mathbf{0}, a\mathbb{V}ar[\hat{\mathbb{E}}[\epsilon(\hat{\theta})]])$$

where $a\mathbb{V}ar[\hat{\theta}] = \frac{1}{T} \cdot (\mathbf{A}\mathbf{G})^{-1} \mathbf{A}\mathbf{S}\mathbf{A}' (\mathbf{A}\mathbf{G})'^{-1}$ is $K \times K$

$a\mathbb{V}ar[\hat{\mathbb{E}}[\epsilon(\hat{\theta})]] = \frac{1}{T} \cdot [\mathbf{I} - \mathbf{G}(\mathbf{A}\mathbf{G})^{-1}\mathbf{A}]\mathbf{S}[\mathbf{I} - \mathbf{G}(\mathbf{A}\mathbf{G})^{-1}\mathbf{A}]'$ is $Q \times Q$

$\mathbf{G} = \partial_{\theta} \mathbb{E}[\epsilon(\theta)]$ is $Q \times K$

$\mathbf{S} = \sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\epsilon_t(\theta)\epsilon_{t-\ell}(\theta)']$ is $Q \times Q$

Estimating G and S

$$G = \partial_{\theta} \mathbb{E}[\epsilon(\theta)] \quad S = \sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\epsilon_t(\theta) \epsilon_{t-\ell}(\theta)']$$

- We need estimates for G and S to estimate $a\mathbb{V}ar[\hat{\theta}]$ and $a\mathbb{V}ar[\hat{\mathbb{E}}[\epsilon(\hat{\theta})]]$
- Estimating G is straightforward: $\hat{G} = \partial_{\theta} \hat{\mathbb{E}}[\epsilon(\hat{\theta})]$
- Estimating S is more nuanced
 - Note: $\mathbb{E}[\epsilon_t(\theta) \epsilon_{t-\ell}(\theta)'] = \mathbb{E}[(\epsilon_t(\theta) - \mathbb{E}[\epsilon(\theta)])(\epsilon_{t-\ell}(\theta) - \mathbb{E}[\epsilon(\theta)])']$ since $\mathbb{E}[\epsilon(\theta)] = \mathbf{0}$ (subtracting $\mathbb{E}[\epsilon(\theta)]$ is not needed but improves finite sample properties)
 - An infeasible HAC (heteroskedasticity and autocorrelation consistent) estimator is

$$\hat{S}_{I\text{HAC}} = \hat{\Omega}_0 + \sum_{\ell=1}^{\ell=\infty} (\hat{\Omega}_{\ell} + \hat{\Omega}_{\ell}') \\ \hat{\Omega}_{\ell} = \hat{\mathbb{E}}[(\epsilon_t(\hat{\theta}) - \hat{\mathbb{E}}[\epsilon(\hat{\theta})])(\epsilon_{t-\ell}(\hat{\theta}) - \hat{\mathbb{E}}[\epsilon(\hat{\theta})])']$$

- A feasible HAC estimator downweights lags (in finite samples) (Andrews (1991))

$$\hat{S}_{\text{HAC}} = \hat{\Omega}_0 + \sum_{\ell=1}^{\ell=L} \omega_{\ell} \cdot (\hat{\Omega}_{\ell} + \hat{\Omega}_{\ell}')$$

- $\omega_{\ell} = 0$ leads to a HC (heteroskedasticity consistent) estimator (White (1980))
- $\omega_{\ell} = 1 - \ell/(L+1)$ leads to a common HAC estimator (Newey, West (1987))

* We can select $L = \text{const} \cdot T^{1/3}$ with optimal const (Newey, West (1994))

- $\omega_{\ell} = 1$ and $L = \text{Overlap}$ deals with overlapping data (Hansen, Hodrick (1980))

The Efficient GMM: Choosing $\hat{A}$ (or $\hat{W}$)

- We have seen that the GMM estimator is based on $\hat{A} \hat{\mathbb{E}}[\epsilon(\hat{\theta})] = \mathbf{0}$
- But how to select $\hat{A}$ (the linear combination of moments to set to zero)?
- The typical approach is to consider the GMM minimization problem:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \quad \hat{\mathbb{E}}[\epsilon(\theta)]' \hat{W} \hat{\mathbb{E}}[\epsilon(\theta)]$$

where $\hat{W} \xrightarrow{P} W$ is a weighting matrix (a $Q \times Q$ and positive definite matrix)

- Minimization Condition: $\hat{G}' \hat{W} \hat{\mathbb{E}}[\epsilon(\hat{\theta})] = \mathbf{0} \Rightarrow$ this is a GMM w/ $\boxed{\hat{A} = \hat{G}' \hat{W}}$
- So, all the GMM results apply with $\hat{A} = \hat{G}' \hat{W} \xrightarrow{P} G' W = A$
- The GMM minimization always has a solution (GMM w/ arbitrary $\hat{A}$ may not)
- So, we typically use this version of the GMM (choose $\hat{W}$ instead of choosing $\hat{A}$)
- It turns out that $\boxed{W = S^{-1}}$ is the (asymptotically) efficient W (Hansen (1982))
 - It results in the lowest $a\mathbb{V}ar[\hat{\theta}]$
 - The GMM asymptotic variance-covariance matrices simplify to:

$$a\mathbb{V}ar[\hat{\theta}] = \frac{1}{T} \cdot (G' S^{-1} G)^{-1} \quad \text{is } K \times K$$

$$a\mathbb{V}ar[\hat{\mathbb{E}}[\epsilon(\hat{\theta})]] = \frac{1}{T} \cdot [S - G(G' S^{-1} G)^{-1} G'] \quad \text{is } Q \times Q$$

- So, (asymptotic) efficiency requires choosing $\hat{W} = \hat{S}^{-1} \xrightarrow{P} S^{-1}$
- Difficulty: any such a $\hat{S}$ requires a $\hat{\theta} \xrightarrow{P} \theta$

The Feasible Efficient GMM

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \quad \widehat{\mathbb{E}}[\epsilon(\theta)]' \widehat{W} \widehat{\mathbb{E}}[\epsilon(\theta)]$$

- The (asymptotically) efficient GMM requires $\widehat{W} = \widehat{S}^{-1} \xrightarrow{P} S^{-1}$
- But such a $\widehat{S}$ requires a $\hat{\theta} \xrightarrow{P} \theta$
- There are many solutions in the literature (but three are most common)

1) Two-Step GMM (Hansen (1982))

- Use an initial arbitrary $\widehat{W}$ (like $\widehat{W} = I$) to obtain the first-step GMM estimate $\hat{\theta}_1$
- Estimate $\widehat{S}$ based on this $\hat{\theta}_1$ (for example, using $\widehat{S}_{HAC}$)
- Set $\widehat{W} = \widehat{S}^{-1}$ and obtain the second-step GMM estimate $\hat{\theta}_2$
- $\hat{\theta}_2$ is a feasible efficient GMM (same asymptotic properties as the efficient GMM)
- Main limitation: in finite samples $\hat{\theta}_2$ depends on the initial $\widehat{W}$ used

2) Iterative GMM (Ferson, Foerster (1994))

- Follow the procedure in (1) until $\hat{\theta}_{h+1} = \hat{\theta}_h$

3) Continuously Updated GMM (Hansen, Heaton, Yaron (1996))

- Use a single step that recognizes the dependency of $\widehat{S}$ on θ :

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \quad \widehat{\mathbb{E}}[\epsilon(\theta)]' \widehat{S}(\theta)^{-1} \widehat{\mathbb{E}}[\epsilon(\theta)]$$

- $\hat{\theta}$ is a feasible efficient GMM (same asymptotic properties as the efficient GMM)
- Main limitation: this optimization is often hard to solve numerically

Important Example: OLS as a GMM

- Many well-known estimators (like OLS and 2SLS) are special cases of GMM (see Hayashi (2000) for an econometrics textbook w/ GMM as its unifying framework)
- As an example, we now show the mapping from GMM to OLS
- Consider the linear regression model ($\mathbf{x}_t$ is a $K \times 1$ vector including a constant)

$$y_t = \mathbf{x}_t' \boldsymbol{\theta} + \varepsilon_t \quad \text{w/ } \mathbb{E}[\mathbf{x}_t \varepsilon_t] = \mathbf{0}$$

- Instead of $\min \sum_{t=1}^T \varepsilon_t^2$ (as in OLS), we use the moment conditions $\mathbb{E}[\mathbf{x}_t \varepsilon_t] = \mathbf{0}$
 - Letting $\epsilon_t(\boldsymbol{\theta}) = \mathbf{x}_t \varepsilon_t = \mathbf{x}_t (y_t - \mathbf{x}_t' \boldsymbol{\theta})$, our GMM estimator solves $\widehat{\mathbf{A}} \widehat{\mathbb{E}}[\epsilon(\widehat{\boldsymbol{\theta}})] = \mathbf{0}$
 - Use $\widehat{\mathbf{A}} = \widehat{\mathbf{G}}' \mathbf{S}^{-1}$ (but any invertible $\widehat{\mathbf{A}}$ yields the same $\widehat{\boldsymbol{\theta}}$ in “just identified” GMM)
 - Solving for $\widehat{\boldsymbol{\theta}}$ yields the GMM estimator (same as the OLS estimator):

$$\mathbf{0} = \widehat{\mathbf{A}} \widehat{\mathbb{E}}[\epsilon(\widehat{\boldsymbol{\theta}})] = \widehat{\mathbb{E}}[\epsilon(\widehat{\boldsymbol{\theta}})] = \widehat{\mathbb{E}}[\mathbf{x}_t (y_t - \mathbf{x}_t' \widehat{\boldsymbol{\theta}})] \Rightarrow \widehat{\boldsymbol{\theta}} = \widehat{\mathbb{E}}[\mathbf{x}_t \mathbf{x}_t']^{-1} \widehat{\mathbb{E}}[\mathbf{x}_t y_t]$$

- We can also estimate $\widehat{\boldsymbol{\theta}}$ standard errors using the efficient GMM asymptotics

$$a\text{Var}[\widehat{\boldsymbol{\theta}}] = \frac{1}{T} \cdot (\mathbf{G}' \mathbf{S}^{-1} \mathbf{G})^{-1} = \frac{1}{T} \cdot \mathbb{E}[\mathbf{x}_t \mathbf{x}_t']^{-1} \left(\sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\varepsilon_t \mathbf{x}_t \mathbf{x}_{t-\ell}' \varepsilon_{t-\ell}] \right) \mathbb{E}[\mathbf{x}_t \mathbf{x}_t']^{-1}$$

$$\mathbf{G} = \partial_{\boldsymbol{\theta}} \mathbb{E}[\epsilon(\boldsymbol{\theta})] = -\mathbb{E}[\mathbf{x}_t \mathbf{x}_t']$$

$$\mathbf{S} = \sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\epsilon_t(\boldsymbol{\theta}) \epsilon_{t-\ell}(\boldsymbol{\theta})'] = \sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\varepsilon_t \mathbf{x}_t \mathbf{x}_{t-\ell}' \varepsilon_{t-\ell}]$$

- We can use the $\widehat{\mathbf{G}} = -\widehat{\mathbb{E}}[\mathbf{x}_t \mathbf{x}_t']$ and $\widehat{\mathbf{S}}_{HAC}$ estimates in place of $\mathbf{G}$ and $\mathbf{S}$

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(1) The SDF

[1.1] The SDF Characterization

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The SDF Equation in GMM Language

- Our SDF equation is (applied to a vector of J asset returns R)

$$\mathbb{E}_t[M_{t+1} \cdot R_{t+1}] = \mathbf{1} \quad \xRightarrow{\text{Law of Iterated Expectations}} \quad \mathbb{E}[M_t \cdot R_t] = \mathbf{1}$$

- Suppose we have a model for the SDF implying $M_t = M(F_t, \theta)$
 - Example: $M_t = \delta \cdot \partial U(C_t, \gamma) / \partial U(C_{t-1}, \gamma)$ implies $F_t = (C_t, C_{t-1})$ and $\theta = (\delta, \gamma)$

- Letting $\epsilon_t(\theta) = M(F_t, \theta) \cdot R_t - \mathbf{1}$, the SDF equation becomes $\mathbb{E}[\epsilon(\theta)] = \mathbf{0}$

- Our GMM estimate for θ solves the system $\widehat{A} \widehat{\mathbb{E}}[\epsilon(\widehat{\theta})] = \mathbf{0}$
- Note that we have $Q = J$ moment conditions
- If $\text{size}(\theta) = K < J$, then the system is “overidentified” and we need to choose $\widehat{A}$
- Typically, $\widehat{\theta} = \underset{\theta}{\operatorname{argmin}} \widehat{\mathbb{E}}[\epsilon(\theta)]' \widehat{W} \widehat{\mathbb{E}}[\epsilon(\theta)] \Rightarrow \widehat{A} = \widehat{G}' \widehat{W}$ so that we have:

$$a\text{Var}[\widehat{\theta}] = \frac{1}{T} \cdot (G'WG)^{-1} G' W S W' G (G'WG)^{-1} \quad \text{is } K \times K$$

$$a\text{Var}[\widehat{\mathbb{E}}[\epsilon(\widehat{\theta})]] = \frac{1}{T} \cdot [I - G(G'WG)^{-1} G' W] S [I - G(G'WG)^{-1} G' W]' \quad \text{is } J \times J$$

- We can use $\widehat{G} = \partial_{\theta} \widehat{\mathbb{E}}[\epsilon(\widehat{\theta})] = \widehat{\mathbb{E}}[\partial_{\theta} M(F_t, \widehat{\theta}) \cdot R_t]$
- $\widehat{S} = \widehat{\mathbb{E}}[\epsilon_t(\widehat{\theta}) \epsilon_t(\widehat{\theta})']$ is valid under the model (because ϵ_t are not autocorrelated)
- But it is better to use $\widehat{S}_{HAC}$ (because it is also valid if the model is wrong)

Choosing $\widehat{W}$ when Estimating the SDF

$$\widehat{\theta} = \underset{\theta}{\operatorname{argmin}} \widehat{\mathbb{E}}[\epsilon(\theta)]' \widehat{W} \widehat{\mathbb{E}}[\epsilon(\theta)] \Rightarrow \boxed{\widehat{G}' \widehat{W} \widehat{\mathbb{E}}[\epsilon(\widehat{\theta})] = \mathbf{0}}$$

- As in any GMM application, $\widehat{W} = \widehat{S}^{-1}$ is the (asymptotically) efficient choice
 - But it may not be the most economically sensible choice
 - We (should) have already chosen a set of J economically relevant assets
 - Reweight pricing errors by $\widehat{W} = \widehat{S}^{-1}$ may emphasize less relevant portfolios
 - For instance, strategies that require large negative positions are infeasible
 - Relatedly, the “bad performance” of a model may be due to one or two assets
 - Finally, $\widehat{S}$ can be a poor estimator for S in small samples (leading to inefficiency)
- A common alternative is to use $\widehat{W} = W = I$
 - It gives each asset the same weight in the pricing error minimization
 - Requires the researcher to choose J economically relevant assets
 - Leads to $\widehat{G}' \widehat{\mathbb{E}}[\epsilon(\widehat{\theta})] = \mathbf{0}$, so it still considers the informativeness of different assets
 - In small samples, it is sometimes more (statistically) efficient than $\widehat{W} = \widehat{S}^{-1}$
- Another alternative is to use $\widehat{W} = \widehat{\mathbb{E}}[R_t R_t']^{-1}$ (Hansen, Jagannathan (1997))
 - Using $W = \mathbb{E}[R_t R_t']$ is equivalent to minimizing distance between $M(F_t, \theta)$ & M_t^*
 - But $\widehat{\mathbb{E}}[R_t R_t']$ can be a poor estimator for $\mathbb{E}[R_t R_t']$ (analogously to $\widehat{S}$)

SDF Equation Pricing Errors

$$\text{SDF Model} \Rightarrow \mathbb{E}[\epsilon(\theta)] = \mathbf{0}$$

- So, $\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]$ are estimated pricing errors
- We can test whether the pricing error of asset j is zero

$$\frac{\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]_j}{\sqrt{a\text{Var}[\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]]_{j,j}}} \stackrel{T \rightarrow \infty}{\sim} \mathcal{N}(0, 1)$$

- We can test whether the pricing errors of group of assets $j \in \mathcal{J}$ are jointly zero

$$\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]'_{\mathcal{J}} \left(a\text{Var}[\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]_{\mathcal{J}}] \right)^{-1} \widehat{\mathbb{E}}[\epsilon(\hat{\theta})]_{\mathcal{J}} \stackrel{T \rightarrow \infty}{\sim} \chi^2_{\text{size}(\mathcal{J})-K}$$

but $(a\text{Var}[\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]_{\mathcal{J}}])^{-1}$ may need a pseudoinverse since $a\text{Var}[\widehat{\mathbb{E}}[\epsilon(\hat{\theta})]]$ is singular

- To interpret the SDF equation pricing errors, note that

$$\begin{aligned} \mathbb{E}[\epsilon_j] &= \mathbb{E}[\mathbf{M} \cdot \mathbf{R}_j - 1] = \mathbb{E}[\mathbf{M}] \cdot \left(\mathbb{E}[\mathbf{R}_j] + \frac{\text{Cov}[\mathbf{M}, \mathbf{R}_j]}{\mathbb{E}[\mathbf{M}]} - \frac{1}{\mathbb{E}[\mathbf{M}]} \right) \\ &\Downarrow \\ \mathbb{E}[\epsilon_j] &= \underbrace{\frac{1}{\mathbb{E}[\mathbf{R}_0]} \cdot \left(\underbrace{\mathbb{E}[\mathbf{R}_j]}_{\text{Expected Return}} - \underbrace{\left(\mathbb{E}[\mathbf{R}_0] + \text{Cov} \left[-\frac{\mathbf{M}}{\mathbb{E}[\mathbf{M}], \mathbf{R}_j} \right] \right)}_{\text{Model Implied Expected Return}} \right)}_{\text{Alpha}} \end{aligned}$$

- $\mathbb{E}[\epsilon_j] > 0 \Rightarrow \mathbb{E}[\mathbf{R}_j]$ is higher than model predicts (and vice versa)

Comparing Nested SDF Models

- Suppose we have two nested models for the SDF
 - The “unrestricted” model: $M(F_t, \theta_u = [\varphi, \phi])$
 - The “restricted” model: $M(F_t, \theta_r = [\varphi, \phi_o])$ w/ ϕ fixed at prespecified ϕ_o
- We can compare these models
 - Estimate the unrestricted model by GMM
 - Test whether ϕ differs from ϕ_o :

$$(\hat{\phi} - \phi_o)' \left(a\mathbb{V}ar[\hat{\phi}] \right)^{-1} (\hat{\phi} - \phi_o) \stackrel{T \rightarrow \infty}{\sim} \chi_{size(\phi)}^2$$

- Note that $a\mathbb{V}ar[\hat{\phi}]$ is a subset of $a\mathbb{V}ar[\hat{\theta}]$ (so, we know how to estimate it)
- If we reject, then the unrestricted model is “preferred” (fits better statistically)
- Otherwise, there is not enough statistical evidence for the unrestricted model
- If we use $\hat{A} = \hat{G}'\hat{W} = \hat{G}'\hat{S}^{-1}$ (the statistically efficient GMM), then

$$T \cdot \left(\mathbb{E}[\epsilon(\hat{\theta}_r)]' \hat{S}(\hat{\theta}_r)^{-1} \mathbb{E}[\epsilon(\hat{\theta}_r)] - \mathbb{E}[\epsilon(\hat{\theta}_u)]' \hat{S}(\hat{\theta}_u)^{-1} \mathbb{E}[\epsilon(\hat{\theta}_u)] \right) \stackrel{T \rightarrow \infty}{\sim} \chi_{size(\phi)}^2$$

- An important limitation of this test is that $\hat{S}(\hat{\theta}_r) \neq \hat{S}(\hat{\theta}_u)$
- So, we may reject the restricted model because $\hat{S}(\hat{\theta}_u)$ is large
- A “better” comparison of pricing errors need to keep $\hat{W}$ fixed across models

Unconditional Implications of the Conditional SDF Equation

- $\mathbb{E}[M_t \cdot R_{j,t}] = 1$ is an unconditional implication of the conditional SDF equation
- But we also have $\mathbb{E}_t[M_{t+1} \cdot (R_{j,t+1} \cdot z_t)] = z_t$ for any instrument z_t
- Interpretation: $R_{j,t+1} \cdot z_t$ is a managed portfolio
 - At time t the investor takes a position of $\$z_t$ in asset j
 - This strategy costs z_t (so the “price” of the managed portfolio is z_t)

- We can now use the law of iterated expectations:

$$\mathbb{E}_t[M_{t+1} \cdot (R_{j,t+1} \cdot z_t)] = z_t \xrightarrow{\text{Law of Iterated Expectations}} \mathbb{E}[(M_{t+1} \cdot R_{j,t+1} - 1) \cdot z_t] = 0$$

- We then have an important result (with $\otimes$ being the Kronecker product)

$$\underbrace{\mathbb{E}_t[M_{t+1} \cdot R_{t+1}] = 1}_{\text{Conditional SDF Equation}} \Leftrightarrow \underbrace{\mathbb{E}[(M_{t+1} \cdot R_{t+1} - 1) \otimes z_t]}_{\text{Unconditional SDF Equation } \forall \text{ Managed Portfolios}} = 0$$

- z_t is the vector of all possible instruments (with $1 \in z_t$ for raw asset returns)
- We can define $e_t(\theta) = (M(F_t, \theta) \cdot R_t - 1) \otimes z_{t-1}$ and apply GMM normally
- Including all possible instruments in z_t is infeasible, so in practice:
 - 1) Estimate the model using sorted portfolios (so BE/ME deciles are implicitly in z_t)
 - 2) and/or choose a small set of instruments and put them into vector z_t
- Note that we still need to model any time-varying parameters in $M(F_t, \theta_t)$

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- We now have a general theory to explain variation in expected returns:

$$\mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}] = 1 \Rightarrow \mathbb{E}_t[R_j] = \mathbb{E}_t[R_0] + \text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1} \right]$$

- And we know how to estimate $M_t = M(F_t, \theta)$ by GMM
- We now explore the linear and log-linear SDFs (common empirically)

[3.1] Linear SDF & Covariance Pricing Models

- We can approximate any SDF model as $M_{t+1} \approx \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1})$
- This implies a linear “covariance pricing model”, $\mathbb{E}_t[xR_j] = b_t' \text{Cov}_t[F, R_j]$
- If we include b_t variation directly in F_t , then $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$
- GMM to estimate $b \Rightarrow$ cross-sectional regression of $\widehat{\mathbb{E}}[xR_j]$ on $\widehat{\text{Cov}}[F, R_j]$
- If F_t are excess returns, then $\widehat{b} = \widehat{\Sigma}_F^{-1} \widehat{\mathbb{E}}[F]$

[3.2] Linear SDF & Beta Pricing Models

- $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$ also implies a linear “beta pricing model”, $\mathbb{E}[xR_j] = \lambda' \beta_j$
- β_j is a vector with the slopes from a multivariate regression of $xR_{j,t}$ on F_t
- If F_t are excess returns, then $\widehat{\lambda} = \widehat{\mathbb{E}}[F]$ (otherwise, regressions of $\widehat{\mathbb{E}}[xR_j]$ on $\widehat{\beta}_j$)
- b reflects a projection of M_t onto F_t while λ_k reflects projection of M_t onto $F_{k,t}$

[3.3] Log-Linear SDF Models

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Linearizing any SDF

- Under No-Arbitrage, we always have a linear SDF: $M_{t+1}^* = \mathbb{E}_t[M^*] \cdot (a_t - xR_{t+1}^*)$
- However, we may be interested in the pricing ability of fundamental factors, F_t
- And we often have a non-linear SDF model, $M_{t+1} = f(F_{t+1}, \theta)$
- There are two main ways to linearize an SDF

1) First Order Taylor Approximation

$$\begin{aligned} M_{t+1} &= f(F_{t+1}, \theta) \approx f(\mathbb{E}_t[F], \theta) + \partial_F f(\mathbb{E}_t[F], \theta)' (F_{t+1} - \mathbb{E}_t[F]) \equiv M_{t+1}^{Linear} \\ M_{t+1}^{Linear} &\Downarrow \\ M_{t+1}^{Linear} &= \mathbb{E}_t[M^{Linear}] \cdot (a_t - b_t' F_{t+1}) \end{aligned}$$

w/ a_t and b_t being time varying parameters (even though θ is not):

$$a_t = 1 + b_t' \mathbb{E}_t[F] \quad \& \quad b_t = -\partial_F f(\mathbb{E}_t[F], \theta) / f(\mathbb{E}_t[F], \theta)$$

2) Stein's lemma ($\text{Cov}[f(x_1), x_2] = \mathbb{E}[\partial f(x_1)] \cdot \text{Cov}[x_1, x_2]$)

- If we assume that F_t and R_t are jointly normal, then Stein's lemma applies:

$$\begin{aligned} 1 &= \mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}] = \mathbb{E}_t[M] \cdot \mathbb{E}_t[R_j] + \text{Cov}_t[f(F_{t+1}, \theta), R_{j,t+1}] \\ &= \mathbb{E}_t[f(F, \theta)] \cdot \mathbb{E}_t[R_j] + \mathbb{E}_t[\partial_F f(F, \theta)]' \cdot \text{Cov}_t[F, R_j] \\ &= \mathbb{E}_t[(\mathbb{E}_t[f(F, \theta)] + \mathbb{E}_t[\partial_F f(F, \theta)]' \cdot (F_{t+1} - \mathbb{E}_t[F])) \cdot R_{j,t+1}] \\ &= \mathbb{E}_t[M_{t+1}^{Linear} \cdot R_{j,t+1}] \end{aligned}$$

so that $M_{t+1}^{Linear} = \mathbb{E}_t[M^{Linear}] \cdot (a_t - b_t' F_{t+1})$ also prices assets

w/ a_t and b_t being time varying parameters (even though θ is not):

$$a_t = 1 + b_t' \mathbb{E}_t[F] \quad \& \quad b_t = -\mathbb{E}_t[\partial_F f(F, \theta)] / \mathbb{E}_t[f(F, \theta)] \quad 3.2$$

Linear SDF $\Leftrightarrow$ Linear Covariance Pricing Model

- If we have the linear SDF model $M_{t+1} = \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1})$, then

$$\mathbb{E}_t[R_j] = \mathbb{E}_t[R_0] + \text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1} \right] = \mathbb{E}_t[R_0] + b_t' \text{Cov}_t[F_{t+1}, R_{j,t+1}]$$

$\Downarrow$

$$\boxed{\mathbb{E}_t[xR_j] = b_t' \text{Cov}_t[F, xR_j]}$$

- This is a “covariance pricing model” ($\text{Cov}_t[F, xR_j]$ drives $\mathbb{E}_t[xR_j]$)
- If the factors (F_t) are excess returns (often the case empirically), then

$$\mathbb{E}_t[F] = \text{Cov}_t[F, F]' b_t \quad \Rightarrow \quad \boxed{b_t = \Sigma_{F,t}^{-1} \mathbb{E}_t[F]}$$

- Note that $M_{t+1}^* = \mathbb{E}_t[M^*] \cdot (a_t - xR_{t+1}^*)$ always holds under No-Arbitrage
- So, our linear SDF model is correct iff $xR_{t+1}^* = b_t' F_{t+1}$
(i.e., $b_t' F_{t+1}$ are excess returns on the maximum Sharpe Ratio portfolio)
- If F_t are not excess returns, we can still construct “factor mimicking portfolios”

- Estimate projection $F_{k,t+1} - \mathbb{E}_t[F_k] = b_{F_k,t}' (xR_{t+1} - \mathbb{E}_t[xR]) + \varepsilon_{t+1}$, which yields

$$b_{F_k,t} = \Sigma_{xR,t}^{-1} \text{Cov}_t[F_k, xR]$$

- Then, $F_{k,t+1}^{proj} = b_{F_k,t}' xR_{t+1}$ is a factor mimicking portfolio
- And $M_{t+1}^{proj} = \mathbb{E}_t[M^{proj}] \cdot (a_t^{proj} + b_t^{proj'} F_{t+1}^{proj})$ implies the same $\mathbb{E}_t[xR]$ as M_t does
- Moreover, our linear SDF model is correct iff $xR_{t+1}^* = b_t^{proj'} F_{t+1}^{proj}$

Linear SDF with Time Varying Parameters

- Our $M_{t+1} = \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1})$ has time varying parameters
- When estimating the model (by GMM) we focus on unconditional implications:

$$\mathbb{E}_t[M_{t+1} \cdot R_{t+1}] = 1 \quad \xRightarrow{\text{Law of Iterated Expectations}} \quad \mathbb{E}[M_t \cdot R_t] = 1$$

- However, the unconditional implication from the model is given by

$$\mathbb{E}[xR_j] = \text{Cov}[b_t' F_{t+1}, xR_{j,t+1}] \neq \mathbb{E}[b_t]' \text{Cov}[F_t, xR_{j,t}]$$

- So, we cannot define $b = \mathbb{E}[b_t]$ and treat the model as $M_t = \mathbb{E}[M] \cdot (a + b' F_t)$
- It we assume joint normality: (Bohrnstedt, Goldberger (1969))

$$\mathbb{E}[xR_j] = \text{Cov}[b_t' F_{t+1}, xR_{j,t+1}] = \mathbb{E}[b_t]' \text{Cov}[F_t, xR_{j,t}] + \mathbb{E}[F_t]' \text{Cov}[b_t, xR_{j,t+1}]$$
 - So, $\text{Cov}[b_t, xR_{j,t+1}] \neq 0$ invalidates $M_t = \mathbb{E}[M] \cdot (a + b' F_t)$
 - Without joint normality third moments also invalidate $M_t = \mathbb{E}[M] \cdot (a + b' F_t)$
- Solution: model $\mathbb{E}_t[M] \cdot a_t$ & $\mathbb{E}_t[M] \cdot b_t$ as linear functions of state variables (s_t)

$$\begin{aligned} M_{t+1} &= \mathbb{E}_t[M] \cdot (a_t + b_t' F_{t+1}) \\ &= \mathbb{E}[M] \cdot ((a_0 + a_1' s_t) + (b_0 + B_1 s_t)' F_{t+1}) \\ &= \mathbb{E}[M] \cdot (a_0 + a_1' s_t + b_0' F_{t+1} + b_1' (s_t \otimes F_{t+1})) \\ &= \mathbb{E}[M] \cdot (a + b' F_{t+1}^{adj}) \quad w/ \quad F_{t+1}^{adj} = [s_t, F_{t+1}, s_t \otimes F_{t+1}] \end{aligned}$$

The Linear SDF GMM Estimator

- Let's estimate a linear SDF equation by GMM
 - Given the prior slide, we consider the linear model $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$ (F_t already captures variation in parameters through s_t)
 - This linear SDF can be written as $M_t = \mathbb{E}[M] \cdot (1 - b' (F_t - \mathbb{E}[F]))$
 - We focus on excess returns (xR) given their prevalence in empirical work
 - We account for conditioning information through managed portfolios
 - And assume that xR_t already includes the relevant managed portfolios ($z_t = 1$)
- So, the (implied unconditional) linear SDF equation is:

$$\mathbb{E}[M_t \cdot xR_t] = 0 \quad \Rightarrow \quad \mathbb{E}[xR_t - b' (F_t - \mathbb{E}[F]) \cdot xR_t] = 0$$

- Defining $\epsilon_t(b) = xR_t - b' (F_t - \mathbb{E}[F]) \cdot xR_t$, we have (for J assets and K factors)

$$G = \partial_b \mathbb{E}[\epsilon(b)] = \mathbb{E}[xR_t (F_t - \mathbb{E}[F])'] = \text{Cov}[F, xR] \quad \text{is } J \times K$$

$$S = \sum_{\ell=-\infty}^{\ell=\infty} \mathbb{E}[\epsilon_t(b) \epsilon_{t-\ell}(b)'] \quad \text{is } J \times J$$

- Our GMM uses $\hat{A} = \hat{G}' \hat{W}$ (i.e., minimizes weighted sum squared pricing errors):

$$\begin{aligned} 0 &= \hat{G}' \hat{W} \hat{\mathbb{E}}[\epsilon_t(\hat{b})] = \hat{G}' \hat{W} \hat{\mathbb{E}}[xR - \hat{b}' (F_t - \hat{\mathbb{E}}[F]) \cdot xR_t] \\ &= \hat{\text{Cov}}[F, xR]' \hat{W} \hat{\mathbb{E}}[xR] - \hat{\text{Cov}}[F, xR]' \hat{W} \hat{\text{Cov}}[F, xR] \hat{b} \end{aligned}$$

$\Downarrow$

$$\hat{b} = \left(\hat{\text{Cov}}[F, xR]' \hat{W} \hat{\text{Cov}}[F, xR] \right)^{-1} \hat{\text{Cov}}[F, xR]' \hat{W} \hat{\mathbb{E}}[xR]$$

Interpreting the Linear SDF GMM Estimator

$$\hat{b} = \left(\hat{\text{Cov}}[F, xR]' \hat{W} \hat{\text{Cov}}[F, xR] \right)^{-1} \hat{\text{Cov}}[F, xR]' \hat{W} \hat{\mathbb{E}}[xR]$$

- $\hat{b}$ has a cross-sectional regression interpretation
 - If $\hat{W} = I$ then $\hat{b}$ is the OLS estimate for a regression of $\hat{\mathbb{E}}[xR_j]$ onto $\hat{\text{Cov}}[F, xR_j]$
 - If $\hat{W} = \hat{S}^{-1}$ then $\hat{b}$ is the GLS estimate for a regression of $\hat{\mathbb{E}}[xR_j]$ onto $\hat{\text{Cov}}[F, xR_j]$
 - And the GMM $a\text{Var}[\hat{b}]$ recognizes that $\hat{\mathbb{E}}[xR_j]$ and $\hat{\text{Cov}}[F, xR_j]$ are estimated
 - But we need to add $\mathbb{E}[F - \mu_F] = 0$ for $a\text{Var}[\hat{b}]$ to recognize the estimate $\hat{\mathbb{E}}[F]$ (this has little effect empirically, but see Cochrane (2005) Chapter 13.2)
- If F are excess returns included in xR and we use $\hat{W} = \hat{S}^{-1}$
 - In this case, the GMM will ignore all other assets in xR (Nagel (2013), Section 3)
 - We effectively have a “just identified” system (K parameters and K factors)
 - The reason is that $\epsilon_{F,t}(b) = 0$ since we can perfectly price the factors
 - So, from a statistical perspective their pricing errors are the most informative
 - In this case, the GMM estimator becomes

$$\hat{b} = (\hat{\text{Cov}}[F, F]' \hat{\text{Cov}}[F, F])^{-1} \hat{\text{Cov}}[F, F]' \hat{\mathbb{E}}[F] = \hat{\Sigma}_F^{-1} \hat{\mathbb{E}}[F]$$

- This estimate makes a lot of sense since we saw that $b = \Sigma_F^{-1} \mathbb{E}[F]$
- $\hat{b}$ is the projection of M_t onto F_t even if $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$ is misspecified (other estimators do not have this property, see Chabi-Yo, Gonçalves, Loudis (2025))

Inference for the Linear SDF GMM Estimator

- We have the standard GMM results for asymptotic inference:

$$\hat{\mathbf{b}} \xrightarrow{T \rightarrow \infty} \mathcal{N}(\mathbf{b}, \mathbf{aVar}[\hat{\mathbf{b}}])$$

$$\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})] \xrightarrow{T \rightarrow \infty} \mathcal{N}(\mathbf{0}, \mathbf{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})])$$

where $\mathbf{aVar}[\hat{\mathbf{b}}] = \frac{1}{T} \cdot (\mathbf{G}' \mathbf{W} \mathbf{G})^{-1} \mathbf{G}' \mathbf{W} \mathbf{S} \mathbf{W}' \mathbf{G} (\mathbf{G}' \mathbf{W} \mathbf{G})'^{-1}$ is $K \times K$

$$\mathbf{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]] = \frac{1}{T} \cdot [\mathbf{I} - \mathbf{G} (\mathbf{G}' \mathbf{W} \mathbf{G})^{-1} \mathbf{G}' \mathbf{W}] \mathbf{S} [\mathbf{I} - \mathbf{G} (\mathbf{G}' \mathbf{W} \mathbf{G})^{-1} \mathbf{G}' \mathbf{W}]' \text{ is } J \times J$$

- Moreover, the asymptotic covariance matrices simplify if $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$

$$\mathbf{aVar}[\hat{\mathbf{b}}] = \frac{1}{T} \cdot (\mathbf{G}' \mathbf{S}^{-1} \mathbf{G})^{-1} \text{ is } K \times K$$

$$\mathbf{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]] = \frac{1}{T} \cdot [\mathbf{S} - \mathbf{G} (\mathbf{G}' \mathbf{S}^{-1} \mathbf{G})^{-1} \mathbf{G}'] \text{ is } J \times J$$

- We can test whether factor $\mathbf{F}_k$ (or group of factors $k \in \mathcal{K}$) helps pricing assets

$$\hat{\mathbf{b}}_k / \sqrt{\mathbf{aVar}[\hat{\mathbf{b}}]_{k,k}} \xrightarrow{T \rightarrow \infty} \mathcal{N}(0, 1)$$

$$\hat{\mathbf{b}}_{\mathcal{K}}' (\mathbf{aVar}[\hat{\mathbf{b}}])_{\mathcal{K}}^{-1} \hat{\mathbf{b}}_{\mathcal{K}} \xrightarrow{T \rightarrow \infty} \chi^2_{\text{size}(\mathcal{K})}$$

- We can test whether the pricing error of asset j (or group of assets $j \in \mathcal{J}$) is zero

$$\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]_j / \sqrt{\mathbf{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]]_{j,j}} \xrightarrow{T \rightarrow \infty} \mathcal{N}(0, 1)$$

$$\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]'_{\mathcal{J}} (\mathbf{aVar}[\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]]_{\mathcal{J}})^{-1} \widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}})]_{\mathcal{J}} \xrightarrow{T \rightarrow \infty} \chi^2_{\text{size}(\mathcal{J}) - K}$$

- Finally, if we use $\widehat{\mathbf{W}} = \widehat{\mathbf{S}}^{-1}$ then we can compare two linear models using

$$T \cdot \left(\widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}}_r)]' \widehat{\mathbf{S}}(\hat{\mathbf{b}}_r)^{-1} \widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}}_r)] - \widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}}_u)]' \widehat{\mathbf{S}}(\hat{\mathbf{b}}_u)^{-1} \widehat{\mathbb{E}}[\epsilon(\hat{\mathbf{b}}_u)] \right) \xrightarrow{T \rightarrow \infty} \chi^2_{\text{size}(\mathbf{b}_u) - \text{size}(\mathbf{b}_r)} \quad \mathbf{3.7}$$

Outline

(1) The SDF

[1.1] The SDF Characterization

[1.2] The SDF & Risk Adjustments

(2) General SDF Estimation

[2.1] The Generalized Method of Moments (GMM)

[2.2] Estimating the SDF by GMM

(3) Linear & Log-Linear SDF Models

[3.1] Linear SDF & Covariance Pricing Models

[3.2] Linear SDF & Beta Pricing Models

[3.3] Log-Linear SDF Models

Linear SDF $\Leftrightarrow$ Linear Beta Pricing Model

- If we have the linear SDF model $M_{t+1} = \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1})$, then

$$\begin{aligned} \mathbb{E}_t[R_j] &= \mathbb{E}_t[R_0] + \text{Cov}_t \left[-\frac{M_{t+1}}{\mathbb{E}_t[M]}, R_{j,t+1} \right] = \mathbb{E}_t[R_0] + b_t' \text{Cov}_t[F_{t+1}, R_{j,t+1}] \\ &= \mathbb{E}_t[R_0] + \underbrace{b_t' \Sigma_{F,t}}_{\lambda_t'} \underbrace{\Sigma_{F,t}^{-1} \text{Cov}_t[F_{t+1}, R_{j,t+1}]}_{\text{Multivariate Regression } \beta_{j,t}} \end{aligned}$$

$\Downarrow$

$$\boxed{\mathbb{E}_t[xR_j] = \lambda_t' \beta_{j,t}}$$

- This is a “beta pricing model” ($\beta_{j,t}$ drives $\mathbb{E}_t[xR_j]$)
- If the factors (F_t) are excess returns (often the case empirically), then

$$\mathbb{E}_t[F] = \beta_{F,t}' \lambda_t = I' \lambda_t \quad \Rightarrow \quad \boxed{\lambda_t = \mathbb{E}_t[F]}$$

- And our linear SDF model is correct iff $xR_{t+1}^* = b_t' F_{t+1}$ (not $xR_{t+1}^* = \lambda_t' F_{t+1}$)
- If F_t are not excess returns, we can still construct F_t^{proj} and then $\lambda_t = \mathbb{E}_t[F^{proj}]$
- Interpreting b_t and λ_t (see Cochrane (2005) Chapter 13.4)
 - b_t elements reflect coefficients from a projection of M_t onto F_t
 - * $b_{k,t} \neq 0$ implies $F_{k,t}$ helps pricing assets controlling for other factors in F_t
 - λ_t elements reflect coefficients from projections of M_t onto each F_t individually
 - * $\lambda_{k,t} \neq 0$ implies $F_{k,t}$ covaries with the SDF (we say “ $F_{k,t}$ is priced”)
 - * Whether $F_{k,t}$ helps pricing assets (controlling for F_t) will depend on $\beta_{j,t}$
 - * To test whether $F_{k,t}$ helps pricing assets (controlling for F_t) just focus on $b_{k,t}$

Beta Pricing Models w/ Time Series Regressions

- Consider the same setup used in our linear SDF GMM estimation
 - Linear SDF model $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$
(F_t already captures variation in parameters through s_t)
 - Focus on excess returns (xR) and assume they include all managed portfolios
- So, the (implied) unconditional beta pricing model is:

$$\mathbb{E}[M_t \cdot xR_t] = 0 \quad \Rightarrow \quad \mathbb{E}[xR_t] = \lambda' \beta_j$$

- Assuming F_t are excess returns (so that $\lambda = \mathbb{E}[F]$), we have $\mathbb{E}[xR_t] = \beta_j' \mathbb{E}[F]$
- Time series regression: $xR_{j,t} = \alpha_j + \beta_j' F_t + \varepsilon_{j,t} \Rightarrow \mathbb{E}[xR_t] = \alpha_j + \beta_j' \mathbb{E}[F]$
- So, the restriction of our beta pricing model is $\alpha_j = 0$ (α reflect pricing errors)
- We can easily obtain $\hat{\lambda} = \hat{\mathbb{E}}[F]$ as well as $\hat{\alpha}_j$ and $\hat{\beta}_j$ (from an OLS regression)
- We can test whether $\alpha_j = 0$ (w/ Newey, West (1987, 1994) standard errors)
- We can test whether a vector of α is jointly zero (Gibbons, Ross, Shanken (1989))
 - Obtain each $\hat{\alpha}_j$ by OLS and assume $\varepsilon_{j,t} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$ (w/ arbitrary $\text{Cor}[\varepsilon_{j1}, \varepsilon_{j2}]$)
 - Then $GRS = \frac{T-J-K}{J} \cdot (1 + \hat{\mathbb{E}}[F]' \hat{\Sigma}_F^{-1} \hat{\mathbb{E}}[F]) \cdot \hat{\alpha}' \hat{\Sigma}_\varepsilon^{-1} \hat{\alpha} \sim \mathcal{F}_{J, T-J-K}$
- And we can even remove the $\varepsilon_{j,t} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$ assumption
 - Estimate the OLS panel $xR_t = \alpha + \beta F_t + \varepsilon_t$ & $a\text{Var}[\hat{\alpha}]$ (w/ Driscoll, Kraay (1998))
 - Perform a Wald test: $\hat{\alpha}' a\text{Var}[\hat{\alpha}] \hat{\alpha} \stackrel{T \rightarrow \infty}{\rightsquigarrow} \chi_{size(\alpha)}^2$

Beta Pricing Models w/ Cross Sectional Regressions

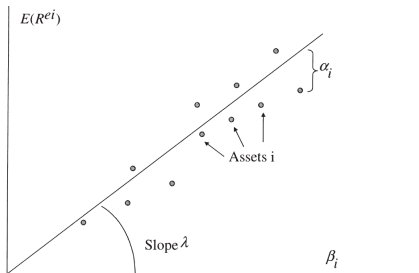
- Consider the same setup used in our linear SDF GMM estimation
 - Linear SDF model $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$
(F_t already captures variation in parameters through s_t)
 - Focus on excess returns (xR) and assume they include all managed portfolios
- So, the (implied) unconditional beta pricing model is:

$$\mathbb{E}[M_t \cdot xR_t] = 0 \quad \Rightarrow \quad \mathbb{E}[xR_j] = \lambda' \beta_j$$

- If F_t are not excess returns (so that $\lambda \neq \mathbb{E}[F]$), then we can either
 - Obtain F_t^{proj} and use it for time series regressions (as in the prior slide)
 - Estimate the model (including λ) using cross sectional regressions
- Implementing cross sectional regressions:

- Obtain $\hat{\beta}_j$ from times series regressions: $xR_{j,t} = \beta_{0,j} + \beta_j' F_t + \varepsilon_{j,t}$

- Obtain $\hat{\lambda}$ from a (no intercept) cross sectional regression: $\mathbb{E}[xR_j] = \lambda' \hat{\beta}_j + \alpha_j$



Beta Pricing Models w/ Cross Sectional Regressions

- Consider the same setup used in our linear SDF GMM estimation
 - Linear SDF model $M_t = \mathbb{E}[M] \cdot (a - b' F_t)$
(F_t already captures variation in parameters through s_t)
 - Focus on excess returns (xR) and assume they include all managed portfolios
- So, the (implied) unconditional beta pricing model is:

$$\mathbb{E}[M_t \cdot xR_t] = 0 \quad \Rightarrow \quad \mathbb{E}[xR_j] = \lambda' \beta_j$$

- If F_t are not excess returns (so that $\lambda \neq \mathbb{E}[F]$), then we can either
 - Obtain F_t^{proj} and use it for time series regressions (as in the prior slide)
 - Estimate the model (including λ) using cross sectional regressions
- Implementing cross sectional regressions:

- Obtain $\hat{\beta}_j$ from times series regressions:
$$xR_{j,t} = \beta_{0,j} + \beta_j' F_t + \varepsilon_{j,t}$$

- Obtain $\hat{\lambda}$ from a (no intercept) cross sectional regression:
$$\widehat{\mathbb{E}}[xR_j] = \lambda' \hat{\beta}_j + \alpha_j$$

* We can add an intercept (λ_0) but model predicts $\lambda_0 = 0$

- Model predicts $\alpha_j = 0$ (α s are the residuals from the cross sectional regression)
- We can estimate the cross sectional regression by OLS or GLS
- OLS & GLS both have close-form $aVar[\hat{\lambda}]$ & $aVar[\hat{\alpha}]$ (Cochrane (2005), Ch. 12.2)
- If $F \in xR$ are excess returns then GLS yields $\hat{\lambda} = \widehat{\mathbb{E}}[F]$ & $\hat{\alpha}_j = \widehat{\mathbb{E}}[xR_j] - \hat{\beta}_j' \widehat{\mathbb{E}}[F]$

Beta Pricing Models w/ Fama-MacBeth Regressions

- Consider a generalization of the setup used in our linear SDF GMM estimation
 - Linear SDF model $M_{t+1} = \mathbb{E}_t[M] \cdot (a_t - b_t' F_{t+1})$
 (F_{t+1} has the raw factors so it does not capture variation in parameters through s_t)
 - Focus on excess returns (xR) which may or may not include managed portfolios

- So, the (implied) conditional beta pricing model is:

$$\mathbb{E}_t[M_{t+1} \cdot xR_{t+1}] = 0 \quad \Rightarrow \quad \mathbb{E}_t[xR_j] = \lambda_t' \beta_{j,t}$$

- F_t may or may not be excess returns
- We can estimate $\mathbb{E}[\lambda_t]$ and $\mathbb{E}[\alpha_{j,t}]$ using Fama-MacBeth regressions
 - Obtain $\hat{\beta}_{j,t}$ from times series regressions on some (potentially moving) window
 - Obtain $\hat{\lambda}_t$ from cross sectional regressions: $xR_{j,t+1} = \lambda_t' \hat{\beta}_{j,t} + \alpha_{j,t}$
 - * We can add an intercept ($\lambda_{0,t}$) but model predicts $\lambda_{0,t} = 0$
 - At the end of this process, we have time series for $\hat{\lambda}_t$ and $\hat{\alpha}_{j,t}$
 - Use regressions to estimate $\lambda = \mathbb{E}[\lambda_t]$ & $\alpha_j = \mathbb{E}[\alpha_{j,t}]$ (w/ Newey, West (1987, 1994))

$$\hat{\lambda}_{k,t} = \lambda_k + \varepsilon_{k,t}^\lambda \quad \& \quad \hat{\alpha}_{j,t} = \alpha_j + \varepsilon_{j,t}^\alpha$$

- Use panel regressions to estimate/test λ s or α s jointly (w/ Driscoll, Kraay (1998))

$$\hat{\lambda}_t = \lambda + \varepsilon_t^\lambda \quad \& \quad \hat{\alpha}_t = \alpha + \varepsilon_t^\alpha$$

- Standard errors from both methods allow for arbitrary $\text{Cor}[xR_{j1}, xR_{j2}]$

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[3.3] Log-Linear SDF Models

Log-Linear SDF Models

- Under No-Arbitrage, there is a positive SDF so that we have a log-linear SDF
- And some economic models imply $m_{t+1} = a_t - b_t' f_{t+1}$ for fundamental factors f_t
- So, we want to understand how to estimate log-linear SDF models
- In general, we can use GMM on the SDF equation, $\mathbb{E}_t[M \cdot R] = 1$
- However, if returns and the SDF are jointly Log-Normal, then
 - Log-linear SDF model $\Leftrightarrow$ Log-linear Covariance/Beta Pricing Model

$$\mathbb{E}_t[r_j] + 0.5 \cdot \text{Var}_t[r_j] = \mathbb{E}_t[r_0] + 0.5 \cdot \text{Var}_t[r_0] + \text{Cov}_t[-m, r_j]$$

$\Downarrow$

$$\begin{aligned} \mathbb{E}_t[xr_j] + 0.5 \cdot (\text{Var}_t[r_j] - \text{Var}_t[r_i]) &= \text{Cov}_t[-m, xr_j] \\ &= b_t' \text{Cov}_t[f, xr_j] \quad (\text{Covariance Pricing Model}) \\ &= b_t' \Sigma_{f,t} \Sigma_{f,t}^{-1} \text{Cov}_t[f, xr_j] \\ &= \lambda_t' \beta_{j,t} \quad (\text{Beta Pricing Model}) \end{aligned}$$

- In some economic models, $b_t = b$ is a function of preference parameters
- In these models, we can use $\mathbb{E}_t[M \cdot R] = 1 \Rightarrow \mathbb{E}[M \cdot R] = 1$ to obtain

$$\begin{aligned} \mathbb{E}[xr_j] + 0.5 \cdot (\text{Var}[r_j] - \text{Var}[r_i]) &= b' \text{Cov}[f, xr_j] \quad (\text{Covariance Pricing Model}) \\ &= \lambda' \beta_j \quad (\text{Beta Pricing Model}) \end{aligned}$$

- We can estimate b or λ using cross sectional (or Fama-MacBeth) regressions

APPENDIX

The SDF & The Growth-Optimal Portfolio

- Same setup as in the slide for the SDF that is linear in payoffs
 - $n = 1, 2, \dots, N$ states of nature
 - $j = 1, 2, \dots, J < N$ non-redundant assets
- Consider the problem of finding the “Growth-Optimal” portfolio: (Long Jr. (1990))

$$\underset{\omega_t}{\text{Max}} \left(\mathbb{E}_t[\log(R_{t+1}^g)] = \mathbb{E}_t[\log(\omega_t' R_{t+1})] \right) \quad \text{s.t. } \omega_t' \mathbf{1} = 1$$

- R^g outperforms any portfolio w/ increasing probability as inv. horizon increases
- R^g is the optimal portfolio of a log-utility investor (but not of other investors)
- The optimality condition for the “Growth-Optimal” portfolio problem is

$$\mathbb{E}_t \left[\frac{1}{R_{t+1}^g} \cdot R_{t+1} \right] = \mathbf{1} \cdot \eta \quad \text{w/ } \eta = \text{Lagrange Multiplier}$$

- Taking a ω_t weighted average of the optimality condition:

$$\mathbb{E}_t \left[\frac{1}{R_{t+1}^g} \cdot \underbrace{\omega_t' R_{t+1}}_{R_{t+1}^g} \right] = \underbrace{\omega_t' \mathbf{1}}_1 \cdot \eta \quad \Rightarrow \quad \eta = 1$$

- So, we have that $M_{t+1}^g = 1/R_{t+1}^g$ is an SDF $\Leftrightarrow \mathbb{E}_t [M_{t+1}^g \cdot R_{t+1}] = 1$
- M^g is not our tradable M^o (unless we have complete markets)
- If the tradable assets satisfy $R > \mathbf{0}$ (e.g., limited liability) then $M^g > 0$

Log Expected Return Equation (Proof) ▸

- From the law of one price, we have

$$\begin{aligned} 1 &= \mathbb{E}_t[M_{t+1} \cdot R_{j,t+1}] = \mathbb{E}_t[e^{m_{t+1}+r_{j,t+1}}] \\ &= e^{\mathbb{E}_t[m_{t+1}+r_{j,t+1}] + 0.5 \cdot \text{Var}_t[m_{t+1}+r_{j,t+1}]} \\ &\Downarrow \end{aligned}$$

$$\mathbb{E}_t[r_j] = -\mathbb{E}_t[m] - 0.5 \cdot \text{Var}_t[m + r_j]$$

- This equation simplifies further for the zero-beta return

$$\begin{aligned} 1 &= \mathbb{E}_t[M_{t+1} \cdot R_{0,t+1}] = \mathbb{E}_t[M_{t+1}] \cdot \mathbb{E}_t[R_{0,t+1}] = \mathbb{E}_t[e^{m_{t+1}}] \cdot \mathbb{E}_t[e^{r_{0,t+1}}] \\ &= e^{\mathbb{E}_t[m_{t+1}+r_{0,t+1}] + 0.5 \cdot (\text{Var}_t[m_{t+1}] + \text{Var}_t[r_{0,t+1}])} \\ &\Downarrow \end{aligned}$$

$$\mathbb{E}_t[r_0] = -\mathbb{E}_t[m] - 0.5 \cdot (\text{Var}_t[m] + \text{Var}_t[r_0])$$

- Subtracting $\mathbb{E}_t[r_0]$ from $\mathbb{E}_t[r_j]$, we have

$$\begin{aligned} \mathbb{E}_t[r_j] - \mathbb{E}_t[r_0] &= -0.5 \cdot (\text{Var}_t[m + r_j] - \text{Var}_t[m] - \text{Var}_t[r_0]) \\ &= -0.5 \cdot (\text{Var}_t[r_j] - \text{Var}_t[r_0] + 2 \cdot \text{Cov}_t[m, r_j]) \\ &\Downarrow \\ \underbrace{\mathbb{E}_t[r_j] + 0.5 \cdot \text{Var}_t[r_j]}_{\log(\mathbb{E}_t[R_j])} &= \underbrace{\mathbb{E}_t[r_0] + 0.5 \cdot \text{Var}_t[r_0]}_{\log(\mathbb{E}_t[R_0])} + \text{Cov}_t[-m, r_j] \end{aligned}$$

- Note: if a risk-free asset exists, then

$$\mathbb{E}_t[r_0] + 0.5 \cdot \text{Var}_t[r_0] = r_{f,t} = \log(R_{f,t})$$